Unit-01

Computer Graphics and Hardware

1.1 History and Applications of Computer Graphics

Computer Graphics

The term Graphics is derived from the Greek word graphics which means something written. E.g. autograph.

The use of a computer and specialized software is to produce and manipulate pictorial images for purpose of animation, business presentations and scientific research.

Computer graphics is the field of study that focuses on creating and manipulating visual content using computers. It involves using algorithms and software to generate, render and manipulate images, videos and animations.

History of Computer Graphics

The fundamental principle and technique derived in the past are still applicable in today's computer graphics technology and generally will be applicable in the future too.

The evolution of graphics can be stated in the following points.

1) Crude plotting on hardcopy devices such as teletypes and line printers' dates from the early days of computing.
2) The Whirlwind computer developed in 1950 at Massachusetts Institute of Technology (MIT) had computer derived CRT displays for output, both for operator use and for cameras producing hardcopy.
3) The SAGE air-defence system developed in the middle 1950's was the first to use command and control CRT displays console on which operators identified targets with the light pens.
4) The beginning of modern interactive graphics was found in Ivan Sutherland's seminal doctoral work on the sketchpad drawing systems. He developed interaction techniques that used the keyboard and light pen for making choices, pointing, drawing and formulating many other ideas and technique still in use today.

Applications of Computer Graphics

Today at most all application programs even for manipulating text or numerical data uses graphics in user interface for visualizing and manipulating the application specific objects. Graphical interaction has replaced textual interaction with alphanumeric terminal. It is an integral part of all computer user interfaces and is important for visualizing 2D,3D and higher-dimensional objects. Some of the application areas of computer graphics are:

1) Computer Aided Design (CAD)

2) Presentation Graphics 3) Computer Art
4) Education & Training
5) Entertainment
6) Visualization
7) Image Processing
8) Graphical User Interface

1.2 Input Devices: Mouse, Keyboard, Touch Panel, Light Pen, Digitizer, Data Glove, Bar Code, OCR, OMR, MICR

Input hardware or input device is used to feed data or information into a computer system. It is usually used to provide input to the computer so that output is generated. Data input device like keyboard is used to provide additional data to the computers whereas pointing and selection devices like mouse, light pens, touch panels are used to provide visual and indication input to the application.

Keyboard

A keyboard is used to input text, labels, menu selections, and graphics commands. It includes:

- **Cursor-control keys:** For moving the cursor on the screen.
- **Function keys:** For performing common operations quickly.

Mouse

A device used to move the screen cursor and select items.

- **Mechanical Mouse:** Uses rollers to detect movement.
- **Optical Mouse:** Uses light and a special pad to track motion.

Touch Panel

A screen that responds to touch, allowing users to select icons or screen positions with a finger or stylus. It works using optical, electrical, or acoustic sensors.

Light Pen

A pen-shaped tool that detects light from CRT screens to select items or draw directly on the screen. It works by capturing light from the screen and recording the cursor position.

Digitizer

A tool for drawing or selecting points on a screen, often used with a stylus or hand cursor.

- **Applications:** Drawing, painting, or inputting 2D/3D coordinates.

Data Glove

A glove with sensors to detect hand and finger movements.

- **Uses:** Manipulating objects in virtual environments.

- **Output:** 2D or 3D visuals on a screen or headset.

Bar Code Reader (BCR)
A scanner that reads bar codes (patterns of light and dark bars) to identify items like products or books. Commonly used in supermarkets and libraries.

Optical Character Reader (OCR)
A device that scans printed or handwritten text, converts it into digital format, and stores it in a computer. Widely used in banks and airlines for processing documents.

Optical Mark Reader (OMR)
A tool that scans marked answer sheets, forms, or surveys and converts the data into a computer-readable format. Commonly used for exams and surveys.

Magnetic Ink Character Reader (MICR)
A device that reads special characters printed with magnetic ink, commonly used by banks to process cheques.

1.3 Hardcopy Output Devices: Printer, Plotter

Hardcopy output devices are tools that produce permanent, physical copies of digital data on materials like paper or vinyl. These devices are essential for generating tangible outputs such as documents, drawings, or labels. Common examples include **printers** for standard documents and images and **plotters** for large-format designs and technical drawings.

Printer

A printer is a device that produces a physical (hardcopy) output of text and graphics on paper. It is widely used for documents, images, and reports. There are various types of printers:

1. **Inkjet Printer**:
   - Uses liquid ink sprayed onto paper.
   - Provides high-quality output, especially for images.
2. **Laser Printer**:
   - Uses toner and laser technology to print.
   - Known for speed, precision, and cost-effectiveness for large volumes.
3. **Dot Matrix Printer**:
   - Uses pins to strike an ink ribbon, forming characters on paper.
   - Commonly used for invoices and forms.
4. **Thermal Printer**:
   - Uses heat to print on special thermal paper.

- Used in receipts and labels.

Plotter

A plotter is a device designed for printing large-format graphics, such as architectural plans, engineering drawings, and maps.

1. **Pen Plotter**:
   - Uses pens to draw on paper.
   - Produces high-precision vector graphics.
2. **Inkjet Plotter**:
   - Uses inkjet technology for high-quality color prints.
   - Commonly used for posters and banners.
3. **Cutting Plotter**:
   - Cuts materials like vinyl for signs and decals.
   - Combines cutting and printing functions.

1.4 Display Devices: CRT (monochrome and colour), LED, LCD Plasma

1.1 Display Devices

Display devices are essential components used to visually present information from a computer or other systems. They vary in technology, features, and usage. Below is a detailed explanation of common display devices:

1. CRT (Cathode Ray Tube)

The **Cathode Ray Tube (CRT)** is one of the earliest and most widely used display technologies, primarily found in older **TVs** and **computer monitors**. **Monochrome CRT**:

- Displays output in a single colour, such as white, green, or amber.
- Commonly used in earlier computers for basic text and graphics.
- Simple and cost-effective, but limited in capability and quality.

Colour CRT:

- Produces full-colour images using three primary colours: Red, Green, and Blue (RGB).
- Inside the screen, phosphor dots glow when struck by an electron beam, creating images.
- Used in older TVs, computer monitors, and arcade machines.
- Bulky and heavy, with high power consumption compared to modern displays.

2. LED (Light Emitting Diode)

LED (Light Emitting Diode) technology has become the standard for most modern displays, replacing older technologies like **CRT** and **LCD** due to its efficiency, thinner design, and superior image quality.

How It Works:

- Uses tiny LED lights for either backlighting (in LCD screens) or as the main light source.
- Emits bright light efficiently, reducing energy usage.

Advantages:

- Slim and lightweight compared to CRTs.
- Provides higher brightness, better contrast, and vivid colours.
- Widely used in TVs, monitors, outdoor displays, and even digital clocks.

3. LCD (Liquid Crystal Display)

LCD (Liquid Crystal Display) is a popular flat-panel display technology used in a wide range of modern devices, including televisions, computer monitors, smartphones, and tablets. It provides clear images and energy efficiency, making it the preferred choice for most modern electronic displays.**How It Works**:

- Liquid crystals are sandwiched between two glass panels.
- When an electric current passes through, the crystals align to control light, forming images.
- Requires a backlight (LED or fluorescent) for illumination.

Advantages:

- Thin, lightweight, and portable.
- Consumes less power than CRT.
- High resolution and image quality.

Common Applications:

- Laptops, smartphones, TVs, and computer monitors.

3. Plasma Display

Plasma display technology is a type of flat-panel display that is known for producing high-quality images, especially in large screen sizes. Plasma displays were once a popular choice for televisions and large monitors, particularly for home entertainment systems and professional applications.

How It Works:

- Contains thousands of tiny cells filled with ionized gases (plasma).
- When electricity passes through, the gases emit ultraviolet light, which excites phosphors to produce images.

Advantages:

- Excellent colour accuracy and high contrast, especially in dark environments.
- Suitable for large screens due to its vibrant display quality.

Disadvantages:

- Heavier and consumes more power than LED or LCD.
- Not as common today, replaced by LED and OLED displays.

1.5 Architecture of Raster Scan and Random Scan Systems

In computer graphics, two main types of display systems are used to render images on the screen: **Raster Scan** and **Random Scan** systems. Each has a distinct architecture and operates differently based on how the image is generated and displayed.

1. Raster Scan System

A **Raster Scan** system is the most common type of display system used in modern computers and TVs. It displays images by illuminating pixels on the screen in a grid-like pattern.

Architecture of Raster Scan System:

1. **Display Screen**:
    - The display is a **grid of pixels** (small points of light) arranged in rows and columns.
    - It is typically made up of **thousands to millions of pixels**, which work together to form an image.
2. **Frame Buffer (Video Memory)**:
    - The frame buffer stores the pixel data, which is sent to the screen.
    - It holds **digital data** for each pixel's color and intensity.
    - The **frame buffer** contains a **2D array** of pixel values, where each entry corresponds to a pixel on the screen.
    - The data is periodically read from the buffer to refresh the screen image.
3. **Raster Scan Process**:
    - The image is generated by scanning the display screen row by row, from left to right and top to bottom, much like the pattern of a television screen.
    - The **electron beam** in a CRT display (or the corresponding display technology in modern devices) follows this path to illuminate each pixel.
    - As the beam moves, the corresponding pixel color is set based on the data from the **frame buffer**.
4. **Horizontal and Vertical Scanning**:
    - **Horizontal Scanning**: The electron beam starts from the leftmost pixel in a row and moves rightward across the screen, illuminating each pixel.

    - **Vertical Scanning**: After completing one row, the beam moves down to the next row and repeats the horizontal scanning process.
    - This process continues until the beam reaches the bottom-right corner, completing the full screen refresh.
5. **Refresh Rate**:
    - The system continuously refreshes the display at a high rate (e.g., 60 Hz, 120 Hz), ensuring the image is constantly updated to avoid flickering.

Advantages of Raster Scan:

- **Flexibility**: Can display complex images like photographs and movies.
- **Standardization**: Used in most modern displays, including computer monitors, smartphones, and TVs.

Disadvantages of Raster Scan:

- **Resolution**: The resolution depends on the number of pixels in the frame buffer. Higher resolutions require more memory and processing power.
- **Energy consumption**: Continuously refreshing the screen can consume significant energy.

2. Random Scan System

A **Random Scan** system (also known as **Vector Scan**) is used in older computer graphics systems, such as some types of **vector displays**. Instead of scanning the entire screen row by row like in raster systems, it directly draws lines or shapes by tracing the coordinates specified in the image data.

Architecture of Random Scan System:

1. **Display Screen**:
    - The screen is divided into **coordinate positions** where the system can directly place a point or draw a line.
2. **Vector Generator (Electron Beam)**:
    - The random scan system uses an **electron beam** (in a CRT system) that can be directed to any position on the screen based on the **vector data** provided.
    - The electron beam moves directly to a specified point without following a row-by-row path.

  - The beam **draws lines** between specified coordinates, much like a plotter.

3. **Memory**:
   - The system uses memory to store vector data. This data contains the coordinates of points or the endpoints of lines to be drawn.
   - Unlike raster scan systems, there is no need to store the entire image in memory, only the lines and shapes to be drawn.
4. **Vector Drawing Process**:
   - The system receives a set of **vector commands** (start and end points of a line or shape).
   - The electron beam moves directly between the specified coordinates and illuminates the screen at those locations.
   - This results in the drawing of lines and shapes that form the final image.
   - After completing one shape or line, the system moves to the next set of coordinates and repeats the process.
5. **Refresh Rate**:
   - The refresh rate is much lower compared to raster scan systems because only specific parts of the screen are updated as needed.
   - The system may need to redraw the image whenever the user interacts with it or when a new shape is displayed.

Advantages of Random Scan:

- **High Precision**: Ideal for applications requiring sharp lines and high precision, such as technical drawings or engineering graphics.
- **Memory Efficiency**: Since only the coordinates of lines and shapes are stored, the system uses less memory than raster scan systems.
- **No Flicker**: Because only the active areas of the screen are updated, there is no flickering of the entire screen.

Disadvantages of Random Scan:

- **Limited Image Types**: Best suited for line drawings, geometric shapes, and wireframe models. Not suitable for displaying complex images like photographs or full-motion video.
- **Higher Cost**: Typically, more expensive to produce and maintain compared to raster scan systems.

- **Speed**: Can be slower for complex images as the system must trace each vector individually.

Difference between random scan and raster scan systems: -

Raster Scan System	Random scan system
1. Raster system produces jagged lines that are plotted as a discrete point sets.	1. Random system produces smooth line drawings because the CRT follows the line path directly.
2. It is less expensive.	2. It is more expensive
3. Modification is difficult.	3. Modification is easy
4. Resolution is low.	4. Resolution is high.
5. Solid pattern is easy to fill.	5. Solid pattern is difficult to fill.
6. Shadow mask technology is used.	6. Beam penetration technology is used.
7. It uses interlacing	7. It does not used interlacing.
8. The refresh rate is independent of	8. The refresh rate depends directly on